

## ASSESSMENT CO2 AT1

1. A simple perceptron model is used to classify whether a student passes or fails based on two input features, namely study hours and attendance. The weights associated with these inputs are 0.6 and 0.4 respectively, and the bias value is – 0.5. For a given student with study hours equal to 2 and attendance equal to 1, compute the net input to the perceptron and apply the step activation function to determine the final output. Based on the result, conclude whether the student passes or fails. (5 marks)

**Given:**

- Weights:  
 $w_1=0.6$  (study hours)  
 $w_2=0.4$  (attendance)
- Inputs:  $x_1=2$ ,  $x_2=1$
- Bias:  $b=-0.5$

**Solution**

The perceptron net input is:

$$\text{Net Input} = (w_1 \cdot x_1) + (w_2 \cdot x_2) + b$$

$$\text{Net Input} = (0.6 \times 2) + (0.4 \times 1) - 0.5$$

$$\text{Net Input} = 1.2 + 0.4 - 0.5$$

$$\text{Net Input} = 1.1$$

Step function rule:

- If net input  $\geq 0 \rightarrow$  Output = **1 (Pass)**
- If net input  $< 0 \rightarrow$  Output = **0 (Fail)**

Since:

$$1.1 \geq 0$$

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| Output = <b>1 (Pass)</b> |
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2. In a neural network, a single neuron is trained using the backpropagation learning rule. The initial weight is 0.5 and the learning rate is 0.1. For an input value of 2, the actual output produced by the neuron is 0.6, whereas the target output is 1. Calculate the error in the output and determine the weight update using the delta rule. Hence, compute the new updated weight after one iteration of training. (5 Marks)

**Given:**

- Initial weight  $w=0.5$
- Learning rate  $\eta=0.1$
- Input  $x=2$
- Actual output  $y=0.6$
- Target output  $t=1$

**1. Error Calculation**

Error is:

$$\text{Error} = t - y$$

$$\text{Error} = 1 - 0.6$$

$$\text{Error} = 0.4$$

**2. Delta Rule**

The delta rule is:

$$\Delta w = \eta(t - y) \times x$$

$$\Delta w = 0.1 \times 0.4 \times 2$$

$$\Delta w = 0.08$$

**3. Updated Weight**

$$w_{\text{new}} = w + \Delta w$$

$$w_{\text{new}} = 0.5 + 0.08$$

$$w_{\text{new}} = 0.58$$